

MUGHEER ISLAM KHATTAK

Software Engineer | Backend & Full-Stack Systems

Abbottabad, Pakistan | Mugheer.ik@gmail.com | +92-320-5802402 | [Linkedin](#) | [Github](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Backend-focused Software Engineer experienced in designing and deploying production-oriented REST APIs with Python, FastAPI, PostgreSQL, Redis, and Docker. Skilled in multi-tenant architecture, role-based access control, caching, rate limiting, and CI/CD through independent, project-based engineering work, with additional background in SQL, dbt, and analytics engineering.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL

Backend & APIs: FastAPI, REST APIs, SQLAlchemy, Pydantic, Alembic, JWT Authentication, RBAC

Databases: PostgreSQL, Redis, Relational Data Modeling, Migrations, Indexing

Testing & DevOps: Pytest, HTTPX, Ruff, pre-commit, Docker, Docker Compose, Git, GitHub Actions, CI/CD, Linux

Frontend & Data: React, TypeScript | SQL, dbt, ETL/ELT, Power BI, DAX

EXPERIENCE

Independent Software Engineering Practice | *Self-Directed Project-Based Work* | May 2025 – Present

- Designed and built backend systems using Python, FastAPI, PostgreSQL, SQLAlchemy, Redis, and Docker.
- Built authenticated REST APIs with role-based access control, resource ownership, and multi-tenant data isolation.
- Applied automated testing, database migrations, containerization, and CI/CD across multiple complete projects.

SELECTED BACKEND ENGINEERING PROJECTS

TeamFlow — Multi-Tenant Task Management API

Python, FastAPI, PostgreSQL, SQLAlchemy, Docker, GitHub Actions | [github](#)

- Built a multi-tenant REST API for organizations, memberships, projects, and tasks using a modular monolith architecture.
- Implemented role-based authorization (Owner/Admin/Member) with server-side enforcement and tenant-isolated data access.
- Designed PostgreSQL schema with constraints and indexes; validated with automated RBAC and cross-tenant access tests, containerized with Docker and CI.

LinkVault — URL Shortening & Analytics API

Python, FastAPI, PostgreSQL, Redis, Docker, GitHub Actions | [github](#)

- Built a URL-shortening REST API with authenticated management, expiration handling, redirects, and click analytics.
- Implemented Redis caching with PostgreSQL fallback, lowering average redirect latency in local benchmarks; added Redis-backed rate limiting.
- Developed automated tests for authentication, ownership, rate limiting, and failure fallback; containerized and CI-integrated.

Reliable Notes API — Authenticated REST API

Python, FastAPI, PostgreSQL, SQLAlchemy, Docker, GitHub Actions | [github](#)

- Built an authenticated REST API with JWT auth, password hashing, and user-level resource ownership.
- Implemented validation, search, pagination, and note lifecycle operations with Alembic-managed PostgreSQL migrations.
- Developed automated API tests against a dedicated test database; containerized and CI-integrated via GitHub Actions.

EDUCATION

Bachelor of Science in Software Engineering — COMSATS University Islamabad, Abbottabad Campus | 2021 – 2025 | CGPA: 2.96/4.00

CERTIFICATIONS

AWS Cloud Practitioner Essentials (AWS) · IBM DevOps & Software Engineering (IBM) · Google IT Automation with Python (Google) · Google Advanced Data Analytics (Google) · Microsoft Power BI Data Analyst (Microsoft)

ADDITIONAL EXPERIENCE

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- Data & Analytics Projects: Built SQL/dbt transformation pipelines and Power BI dashboards for SaaS revenue, healthcare, and stock market analytics use cases.
 - News Broadcast Updates (Final Year Project, Team of 3): React Native/Firebase mobile news app with AI-assisted video summarization (Python/Flask).

Languages: English (Professional Working Proficiency) · Urdu (Native) · Pashto (Native/Professional)